**LAPORAN JOBSHEET 18: OPTIMIZE MATKUL PEMOGRAMAN BERBASIS NETWORK**
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## PRAKTIKUM 1 – IMAGE OPTIMIZATION

### A. Optimasi Gambar Lokal (Public Folder)

Studi Kasus:

Mengganti tag <img> pada halaman 404 dengan next/image. Langkah:

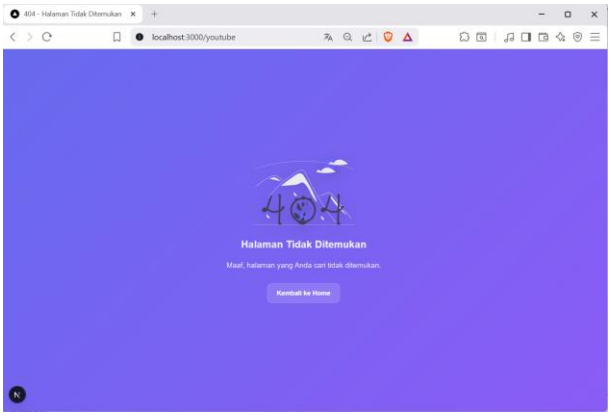
Buka file src/pages/404.tsx

Modifikasi line 7 menjadi line 8-11

pemograman_framework - 404.tsx

```
1  import styles from "@/styles/404.module.scss";
2  import Link from "next/dist/client/link";
3  import Image from "next/image";
4  const Custom404 = () => {
5      return (
6          <div className={styles.error}>
7              <head>
8                  <title>404 - Halaman Tidak Ditemukan</title>
9              </head>
10             {/* <img src="/page-not-found.svg" alt="404" className={styles.error__image} /> */}
11             <Image
12                 src="/page-not-found.svg"
13                 alt="404"
14                 width={400}
15                 height={200}
16                 className={styles.error__image}
17             />
18             <h2 className={styles.error__subtitle}>
19                 Halaman Tidak Ditemukan
20             </h2>
21             <p className={styles.error__desc}>Maaf, halaman yang Anda cari tidak ditemukan.</p>
22             <Link href="/" className={styles.error__button}>Kembali ke Home</Link>
23         </div>
24     );
25 };
26 export default Custom404;
```

Hasil:




- Warning hilang
- Image dioptimasi otomatis
- Mengurangi bandwidth

• Mendukung lazy loading otomatis

**B. Optimasi Gambar Remote (External URL)**

Buka file views/product/index.tsx

Modifikasi file index.tsx

pemograman_framework - index.tsx

```
3  import Image from "next/image";
```

Diset length > 10 karena ada data yg isinya cuma http sehingga muncul error karena image tidak terender.

**Note : dikarenakan gambar diambil dari url tertentu maka konfigurasi berbeda**

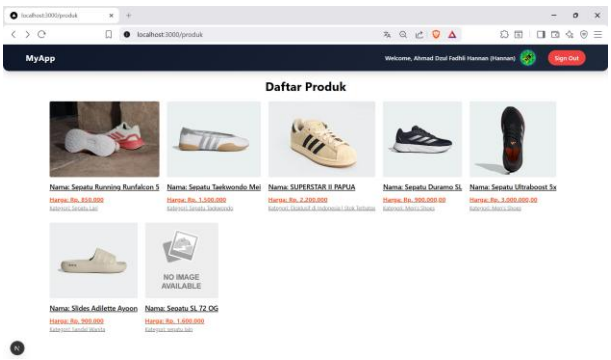
Buka file next.config.js

pemograman_framework - next.config.js

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  reactStrictMode: true,
  images: {
    remotePatterns: [
      {
        protocol: 'https',
        hostname: 'assets.adidas.com',
        port: "",
        pathname: "/**",
      }
    ],
  },
}

module.exports = nextConfig
```

Hasil:



• Gambar di-proxy melalui /_next/image

• Performa lebih optimal

• Kompresi otomatis

**PRAKTIKUM 2 – FONT OPTIMIZATION**

**A. Menggunakan next/font**

Buka file index.tsx pada folder Appshell/index.tsx dan modifkasi

pemograman_framework - index.tsx

```
3  import { Roboto } from "next/font/google";
```

pemograman_framework - index.tsx

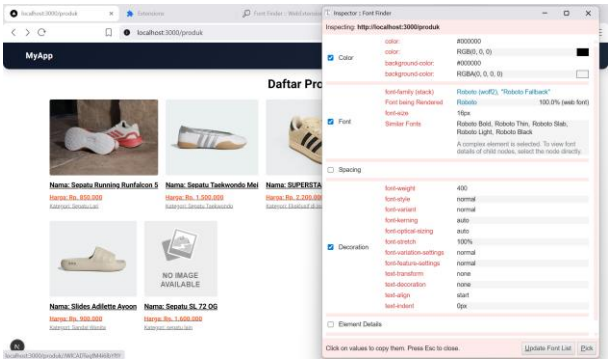
```
11  const roboto = Roboto({
12      subsets: ['latin'],
13      weight: ['400', '500', '700'],
14  });
```

pemograman_framework - index.tsx

```
23  <main className={roboto.className}>
```

Jalankan browser localhost:3000/produk maka font akan berubah menjadi roboto

untuk mengecek fontnya bisa menggunakan extension FontFinder




Hasil:

• Tidak perlu load dari CDN manual

• Tidak blocking render

• Performance meningkat

• Tidak terjadi FOUT (Flash of Unstyled Text)

**PRAKTIKUM 3 – SCRIPT OPTIMIZATION**

**B. Menggunakan next/script**

Buka file index.tsx pada folder layouts/Navbar dan modifikasi

pemograman_framework - index.tsx

```
1  import Script from "next/dist/client/script";
```

pemograman_framework - index.tsx

```
13  {/* <div className={styles.navbar__brand}>
14      MyApp
15  </div> */}
16
17  <div className={styles.navbar__brand} id="title"></div>
18  <Script id="title-script" strategy="lazyOnload">
19      {`document.getElementById('title').innerHTML='MyApp';`}
20  </Script>
```

Pada kasus diatas kita merubah line 11 menggunakan model Typescript dapat terlihat ketika kita refresh web produk tulisan myApp akan terlihat berkedip

Perbedaan mendasar antara Line 11-13 dan Line 15-18 pada file index.tsx Anda terletak pada metode rendering teks dan interaksi dengan DOM (Document Object Model). Berikut adalah rincian perbedaannya:

I. Metode Rendering
   a. Line 11-13 (Standard React/JSX): Ini adalah cara standar React. Teks "MyApp" ditulis langsung di dalam tag div. React akan langsung merender teks ini ke dalam HTML saat komponen dimuat.
   b. Line 15-18 (Next.js Script & Manual DOM): Menggunakan komponen <Script> dari Next.js dengan strategi lazyOnload. Teks tidak ada di dalam HTML saat awal dimuat, melainkan baru "disuntikkan" secara manual menggunakan perintah JavaScript document.getElementById('title').innerHTML = 'MyApp'; setelah script tersebut diunduh di latar belakang.

II. Performa dan SEO
   a. Line 11-13: Sangat baik untuk SEO karena teks "MyApp" langsung terbaca oleh robot pencari (crawler) dalam kode sumber HTML.
   b. Line 15-18: Kurang baik untuk SEO untuk konten penting karena teks baru muncul setelah JavaScript dijalankan. Strategi lazyOnload berarti script ini dijalankan paling

akhir setelah semua sumber daya utama selesai dimuat, sehingga mungkin ada jeda waktu (delay) sebelum teks muncul di layar.

III. Keamanan (XSS)

a. Line 11-13: Aman karena React secara otomatis melakukan escape pada string untuk mencegah serangan Cross-Site Scripting (XSS).
b. Line 15-18: Memiliki risiko keamanan lebih tinggi karena menggunakan .innerHTML. Jika data yang dimasukkan berasal dari input user, ini bisa dimanfaatkan untuk menyuntikkan skrip berbahaya.

**C. Strategi Script**

| Strategy | Fungsi |
|---|---|
| **beforeInteractive** | Sebelum halaman interaktif |
| **afterInteractive** | Setelah halaman interaktif |
| **lazyOnload** | Setelah semua selesai |
| **worker** | Web worker |

Hasil:

• Script tidak blocking

• Cocok untuk Google Analytics

• Performa lebih ringan

**PRAKTIKUM 4 – OPTIMASI AVATAR DENGAN NEXT/IMAGE**

Buka file index.tsx pada folder layouts/navbar dan modifikasi :

pemograman_framework - index.tsx

```
1  import Script from "next/dist/client/script";
```

pemograman_framework - index.tsx

```
27  {data.user.image && (
28      <Image
29          src={data.user.image}
30          alt={data.user.fullname}
31          className={styles.navbar__user__image}
32          width={50}
33          height={50}
34      />
35  )}
```

Tambahkan hostname Google dan Github:

pemograman_framework - next.config.js

```
/** @type {import('next').NextConfig} */
const nextConfig = {
  reactStrictMode: true,
  images: {
    remotePatterns: [
      {
        protocol: 'https',
        hostname: 'assets.adidas.com',
        port: "",
        pathname: "/**",
      },
      {
        protocol: 'https',
        hostname: 'lh3.googleusercontent.com',
        port: "",
        pathname: "/**",
      },
      {
        protocol: 'https',
        hostname: 'avatars.githubusercontent.com',
        port: "",
        pathname: "/**",
      }
    ],
  },
}

module.exports = nextConfig
```

## TUGAS PRAKTIKUM

### 1. Optimasi semua image di project menggunakan next/image

1. src/components/layouts/navbar/index.tsx

pemograman_framework - index.tsx

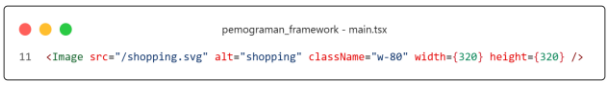
```
{data.user.image && (
    <Image
        src={data.user.image}
        alt={data.user.fullname}
        className={styles.navbar__user__image}
        width={50}
        height={50}
    />
```

2. src/views/lapar/index.tsx

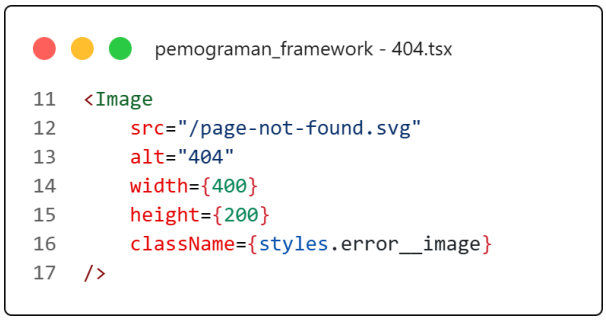
3. src/views/DetailProduct/index.tsx

4. src/views/produk/main.tsx


5. src/views/lapar/lapar/main.tsx



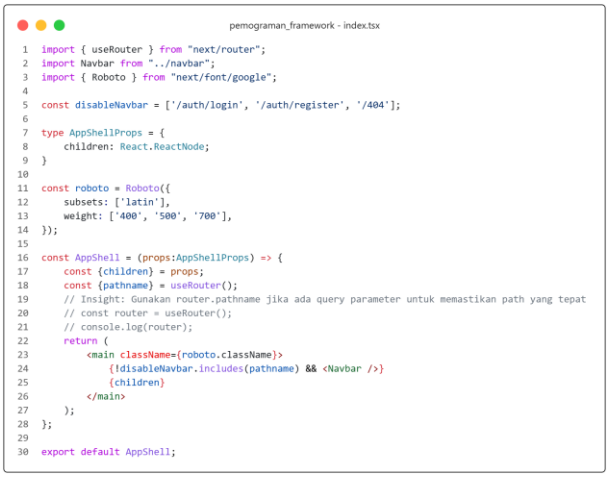

6. src/pages/404.tsx




7. src/views/product/index.tsx

**2. Gunakan minimal 1 font dari next/font**

/Appshell/index.tsx




Menggunakan Roboto

**3. Tambahkan script Google Analytics menggunakan next/script**

Tambahkan /components/analytics/Analytics.tsx

pemograman_framework - Analytics.tsx

```tsx
import Script from "next/script";
import { useEffect } from "react";
import { useRouter } from "next/router";

const gaId = process.env.NEXT_PUBLIC_GA_ID;

declare global {
    interface Window {
        gtag?: (...args: unknown[]) => void;
    }
}

const Analytics = () => {
    const router = useRouter();

    useEffect(() => {
        if (!gaId) return;

        const handleRouteChange = (url: string) => {
            if (typeof window.gtag !== "function") return;
            window.gtag("config", gaId, {
                page_path: url,
            });
        };

        router.events.on("routeChangeComplete", handleRouteChange);
        return () => {
            router.events.off("routeChangeComplete", handleRouteChange);
        };
    }, [router.events]);

    if (!gaId) return null;

    return (
        <>
        <Script
            src={`https://www.googletagmanager.com/gtag/js?id=${gaId}`}
            strategy="afterInteractive"
        />
        <Script id="google-analytics" strategy="afterInteractive">
            {`
                window.dataLayer = window.dataLayer || [];
                function gtag(){dataLayer.push(arguments);}
                gtag('js', new Date());
                gtag('config', '${gaId}', { page_path: window.location.pathname });
            `}
        </Script>
        </>
    )
}

export default Analytics;
```

**4. Terapkan dynamic import pada minimal 1 komponen**

pemograman_framework - _app.tsx

```tsx
import "../styles/globals.css";
import type { AppProps } from 'next/app'
import AppShell from '@/components/layouts/Appshell';
import {SessionProvider} from "next-auth/react";
import dynamic from "next/dynamic";

const Analytics = dynamic(() => import('@/components/analytics/Analytics'), { ssr: false });

export default function App({ Component, pageProps: {session, ...pageProps} }: AppProps) {
  return (
    <SessionProvider session={pageProps.session}>
    <Analytics />
    <AppShell>
      <Component {...pageProps} />
    </AppShell>
    </SessionProvider>

  );
};

```

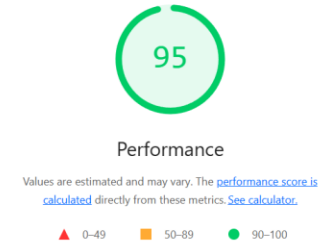
**5. Dokumentasikan perubahan performa (screenshot Lighthouse)**

Note: Untuk perbandingan agar adil:

- pada branch pertemuan-6 sebagai patokan kondisi sebelum optimasi dan branch pertemuan-7 sebagai patokan kondisi setelah optimasi,
- semua ekstensi browser mati,
- route ke /produk,
- untuk platform yg digunakan adalah desktop.
- Pada env.local NEXT_PUBLIC_GA_ID= dihapus

# Sebelum

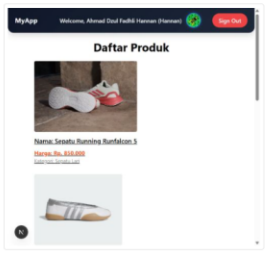
http://localhost:3000/produk




Performance

Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.

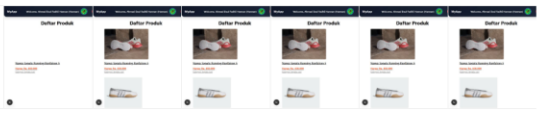
▲ 0–49 ■ 50–89 ● 90–100




METRICS Expand view

| | |
|---|---|
| First Contentful Paint 0.3 s | Largest Contentful Paint 1.4 s |
| Total Blocking Time 0 ms | Cumulative Layout Shift 0.049 |
| Speed Index 1.0 s | |

View Treemap



Show audits relevant to: All FCP LCP CLS

INSIGHTS

- ▲ Improve image delivery — Est savings of 136 KiB
- ▲ Legacy JavaScript — Est savings of 11 KiB
- ▲ LCP request discovery
- ■ Use efficient cache lifetimes — Est savings of 1 KiB
- ○ Layout shift culprits
- ○ LCP breakdown
- ○ 3rd parties

These insights are also available in the Chrome DevTools Performance Panel - record a trace to view more detailed information.

DIAGNOSTICS

- ▲ Minify JavaScript — Est savings of 188 KiB
- ▲ Reduce unused JavaScript — Est savings of 558 KiB
- ▲ Page prevented back/forward cache restoration — 1 failure reason
- ○ User Timing marks and measures — 24 user timings

More information about the performance of your application. These numbers don't directly affect the Performance score.

PASSED AUDITS (18) Show

Captured at Apr 11, 2026, 10:06 AM GMT+7
Emulated Desktop with Lighthouse 13.0.2
Single page session
Initial page load
Custom throttling
Using Chromium 146.0.0.0 with devtools

Generated by **Lighthouse** 13.0.2 | File an issue

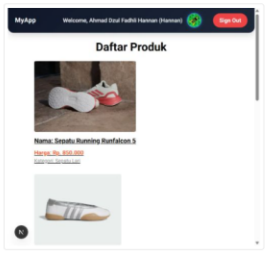
## Sesudah

http://localhost:3000/produk


Performance

Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.

▲ 0–49 ■ 50–89 ● 90–100




METRICS Expand view

| | |
|---|---|
| First Contentful Paint 0.3 s | Largest Contentful Paint 1.3 s |
| Total Blocking Time 0 ms | Cumulative Layout Shift 0.048 |
| Speed Index 1.1 s | |

View Treemap

Show audits relevant to: All FCP LCP TBT CLS

INSIGHTS

- ▲ LCP request discovery
- ■ Use efficient cache lifetimes — Est savings of 1 KiB
- ■ Legacy JavaScript — Est savings of 11 KiB
- ○ Layout shift culprits
- ○ LCP breakdown
- ○ 3rd parties

These insights are also available in the Chrome DevTools Performance Panel - record a trace to view more detailed information.

DIAGNOSTICS

- ▲ Minify JavaScript — Est savings of 196 KiB
- ▲ Reduce unused JavaScript — Est savings of 333 KiB
- ▲ Page prevented back/forward cache restoration — 1 failure reason
- ○ User Timing marks and measures — 26 user timings
- ○ Avoid long main-thread tasks — 1 long task found

More information about the performance of your application. These numbers don't directly affect the Performance score.

PASSED AUDITS (18) Show

Captured at Apr 11, 2026, 10:04 AM GMT+7
Emulated Desktop with Lighthouse 13.0.2
Single page session
Initial page load
Custom throttling
Using Chromium 146.0.0.0 with devtools

Generated by **Lighthouse** 13.0.2 | File an issue

**Kesimpulan**

Optimasi yang dilakukan berhasil meningkatkan performa halaman /produk, meskipun peningkatannya tidak terlalu besar. Skor Lighthouse naik dari 95 menjadi 96.

Beberapa metrik menunjukkan perbaikan:

- LCP turun dari 1.4s menjadi 1.3s
- TTI turun dari 1.4s menjadi 1.3s
- CLS sedikit lebih stabil, dari 0.049 menjadi 0.048
- TBT tetap sangat baik di angka 0 ms

**REFLEKSI & DISKUSI**

1. Mengapa <img> biasa tidak optimal?

Karena gambar yang tidak terkompresi dengan baik atau tidak menggunakan teknik seperti lazy loading dapat memperlambat waktu muat halaman.

2. Apa perbedaan font CDN dan next/font?

Font CDN mengambil file font dari server eksternal saat runtime, sedangkan next/font mengelola, mengoptimasi, dan menyajikan font langsung dari aplikasi agar lebih cepat dan konsisten.

3. Mengapa script bisa membuat website lambat?

Karena menambah beban download, parsing, dan eksekusi JavaScript di main thread saat halaman dimuat.

4. Kapan harus menggunakan dynamic import?

Saat komponen atau library tidak dibutuhkan di initial render agar bundle awal lebih kecil dan waktu muat awal lebih cepat.

5. Apa dampak bundle size terhadap UX?

Bundle size yang besar memperburuk UX karena meningkatkan waktu loading, menghambat interaksi awal, dan membuat aplikasi terasa lambat terutama di perangkat atau jaringan lemah.

**KESIMPULAN**

Dalam praktikum ini mahasiswa telah mempelajari:

✅ Image Optimization

✅ Remote Image Configuration

✅ Font Optimization

✅ Script Optimization

✅ Dynamic Import

✅ Lazy Loading

Semua fitur ini merupakan keunggulan utama Next.js dalam meningkatkan performa aplikasi modern.

**LAPORAN JOBSHEET 19: UNIT TESTING MATKUL PEMOGRAMAN BERBASIS NETWORK**
**AHMAD DZUL FADHLI HANNAN | TI3F | 2341720106**

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## PRAKTIKUM 1 – SETUP JEST DI NEXT.JS

### Langkah 1- Install Dependencies

Jalankan:

npm install jest jest-environment-jsdom @testing-library/react @testing-library/jest-dom --save-dev --force


### Langkah 2 - Buat File Konfigurasi

Doc : https://nextjs.org/docs/pages/guides/testing/jest

Buat file:

o jest.config.mjs

Isi dengan:

pemograman_framework - jest.config.mjs

```
1  import nextJest from 'next/jest.js';
2
3  const createJestConfig = nextJest({
4      dir: './',
5  })
6
7  const config = {
8      coverageProvider: 'v8',
9      testEnvironment: 'jsdom',
10 }
11
12 export default createJestConfig(config);
```

### Langkah 3 - Tambahkan Script di package.json

pemograman_framework - package.json

```
10  "test": "jest --passWithNoTests -u",
11  "test:coverage": "npm run test -- --coverage",
12  "test:watch": "jest --watch -u"
```

**PRAKTIKUM 2 – STRUKTUR FOLDER TESTING**

Buat folder:

src/__test__/

Struktur contoh:

src

├─── pages

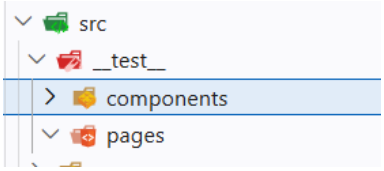
├─── components

├─── views

└─── __test__

├─── pages

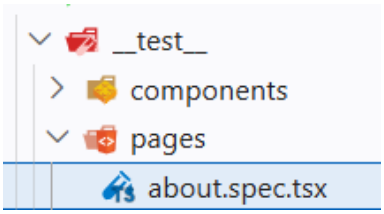
└─── components




**PRAKTIKUM 3 – TESTING HALAMAN ABOUT**

**Langkah 1 - Buat File Testing**

src/__test__/pages/about.spec.tsx



**Langkah 2 - Contoh Testing Snapshot. Pada about.spec.tsx tambahkan code berikut :**

pemograman_framework - about.spec.tsx

```tsx
import { render } from '@testing-library/react';
import AboutPage from "@/pages/about";

describe("AboutPage", () => {
    it("renders about page correctly", () => {
        const page = render(<AboutPage />);
        expect(page).toMatchSnapshot();
    });
})

```

**Langkah 3 - Jalankan Testing**

• npm run test

• Jika berhasil:

• PASS about.spec.tsx

```
> npm run test

> my-app@0.1.0 test
> jest --passWithNoTests -u

 PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders about page correctly (20 ms)

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 passed, 1 total
Time:        0.815 s, estimated 1 s
Ran all test suites.
```

**PRAKTIKUM 4 – COVERAGE REPORT**

Jalankan:

o npm run test:coverage

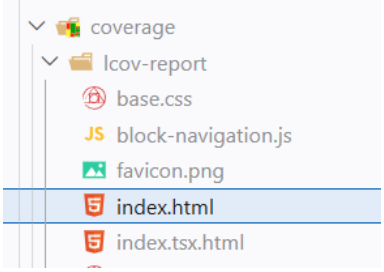
```
> jest --passWithNoTests -u --coverage

 PASS  src/__test__/pages/about.spec.tsx
  AboutPage
    √ renders about page correctly (21 ms)

-----------|---------|----------|---------|---------|-------------------
File       | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------|---------|----------|---------|---------|-------------------
All files  |     100 |      100 |     100 |     100 |
 index.tsx |     100 |      100 |     100 |     100 |
-----------|---------|----------|---------|---------|-------------------
Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 passed, 1 total
Time:        1.673 s
Ran all test suites.
>
```

Akan muncul folder:

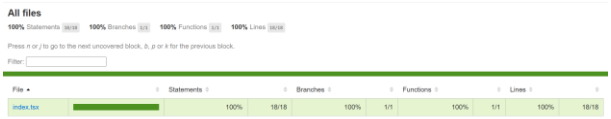
o coverage/

o Buka:

▪ coverage/lcov-report/index.html ( buka di melalui explorer)




Target perusahaan biasanya:




Minimum 80% coverage

**PRAKTIKUM 5 – KONFIGURASI COVERAGE LENGKAP**

Update jest.config.mjs:

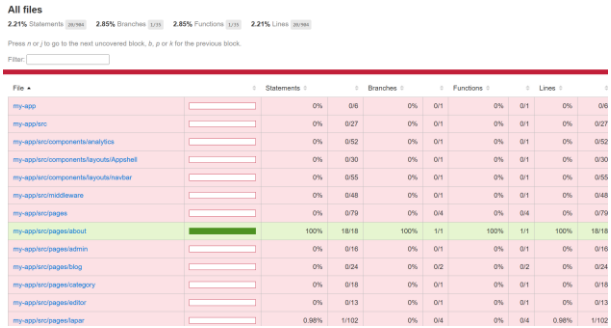
pemograman_framework - jest.config.mjs

```mjs
import nextJest from 'next/jest.js';

const createJestConfig = nextJest({
    dir: './',
})

const config = {
    coverageProvider: 'v8',
    testEnvironment: 'jsdom',
    modulePaths: ['<rootDir>/src/'],
    collectCoverage: true,
    collectCoverageFrom: [
        '**/*.{ts,tsx}',
        '**/*.d.ts',
        '!**/node_modules/**',
        '!**/.next/**',
        '!**/coverage/**',
        '!**/jest.config.mjs',
        '!**/next.config.mjs',
        '!**/types/**',
        '!**/views/**',
        '!**/pages/api/**',
    ],
}

export default createJestConfig(config);
```

Jalankan npm run test:coverage

```
> npm run test:coverage
  index.tsx                                |      0 |        0 |      0 |      0 | 1-30
 my-app/src/components/layouts/navbar      |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-55
 my-app/src/middleware                     |      0 |        0 |      0 |      0 |
  withAuth.ts                              |      0 |        0 |      0 |      0 | 1-48
 my-app/src/pages                          |      0 |        0 |      0 |      0 |
  404.tsx                                  |      0 |        0 |      0 |      0 | 1-26
  _app.tsx                                 |      0 |        0 |      0 |      0 | 1-19
  _document.tsx                            |      0 |        0 |      0 |      0 | 1-13
  index.tsx                                |      0 |        0 |      0 |      0 | 1-21
 my-app/src/pages/about                    |    100 |      100 |    100 |    100 |
  index.tsx                                |    100 |      100 |    100 |    100 |
 my-app/src/pages/admin                    |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-16
 my-app/src/pages/blog                     |      0 |        0 |      0 |      0 |
  [slug].tsx                               |      0 |        0 |      0 |      0 | 1-15
  index.tsx                                |      0 |        0 |      0 |      0 | 1-9
 my-app/src/pages/category                 |      0 |        0 |      0 |      0 |
  [[...slug]].tsx                          |      0 |        0 |      0 |      0 | 1-18
 my-app/src/pages/editor                   |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-13
 my-app/src/pages/lapar                    |   0.98 |        0 |      0 |   0.98 |
  [id].tsx                                 |      0 |        0 |      0 |      0 | 1-17
  index.tsx                                |   3.12 |        0 |      0 |   3.12 | 1-31
  server.tsx                               |      0 |        0 |      0 |      0 | 1-25
  static.tsx                               |      0 |        0 |      0 |      0 | 1-28
 my-app/src/pages/produk                   |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-34
  server.tsx                               |      0 |        0 |      0 |      0 | 1-25
  static.tsx                               |      0 |        0 |      0 |      0 | 1-28
 my-app/src/pages/produk/[produk]          |      1 |        0 |      0 |      1 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-20
  server.tsx                               |      0 |        0 |      0 |      0 | 1-29
  static.tsx                               |   1.96 |        0 |      0 |   1.96 | 1-50
 my-app/src/pages/profil                   |      0 |        0 |      0 |      0 |
  edit.tsx                                 |      0 |        0 |      0 |      0 | 1-14
  index.tsx                                |      0 |        0 |      0 |      0 | 1-13
 my-app/src/pages/setting                  |      0 |        0 |      0 |      0 |
  app.tsx                                  |      0 |        0 |      0 |      0 | 1-9
 my-app/src/pages/shop                     |      0 |        0 |      0 |      0 |
  [[...slug]].tsx                          |      0 |        0 |      0 |      0 | 1-15
 my-app/src/pages/user                     |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-9
 my-app/src/pages/user/password            |      0 |        0 |      0 |      0 |
  index.tsx                                |      0 |        0 |      0 |      0 | 1-9
 my-app/src/utils                          |      0 |        0 |      0 |      0 |
  firebase.ts                              |      0 |        0 |      0 |      0 | 1-19
 my-app/src/utils/db                       |      0 |        0 |      0 |      0 |
  servicefirebase.ts                       |      0 |        0 |      0 |      0 | 1-139
 my-app/src/utils/swr                      |      0 |        0 |      0 |      0 |
  fetcher.ts                               |      0 |        0 |      0 |      0 | 1-2
-------------------------------------------|--------|----------|--------|--------|-----------------
Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 passed, 1 total
Time:        1.664 s
Ran all test suites.
```

Jika dilihat di index.htmlnya




## PRAKTIKUM 6 – TESTING DENGAN GETBYTESTID

Tambahkan pada About Page

- <h1 data-testid="title">About Page</h1>

pemograman_framework - index.tsx

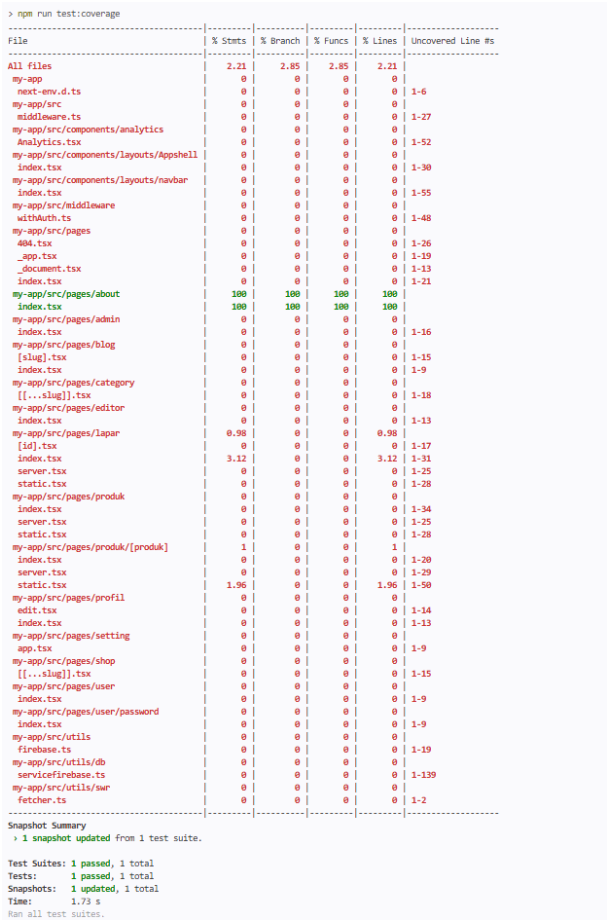
```
11  <h1 data-testid="title">Ini Adalah Halaman About</h1>
```

Update Testing pada about.spec.tsx

pemograman_framework - about.spec.tsx

```
1  import { render } from '@testing-library/react';
2  import AboutPage from "@/pages/about";
3
4  describe("AboutPage", () => {
5      it("renders about page correctly", () => {
6          const page = render(<AboutPage />);
7          expect(page.getByTestId("title").textContent).toBe("Ini Adalah Halaman About");
8          expect(page).toMatchSnapshot();
9      });
10 })
11
```

Dicoba untuk di run



```
> npm run test:coverage

-----------------------------------------|---------|----------|---------|---------|-------------------
File                                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------------------------------------|---------|----------|---------|---------|-------------------
All files                                |    2.21 |     2.85 |    2.85 |    2.21 |
 my-app                                  |       0 |        0 |       0 |       0 |
  next-env.d.ts                          |       0 |        0 |       0 |       0 | 1-6
 my-app/src                              |       0 |        0 |       0 |       0 |
  middleware.ts                          |       0 |        0 |       0 |       0 | 1-27
 my-app/src/components/analytics         |       0 |        0 |       0 |       0 |
  Analytics.tsx                          |       0 |        0 |       0 |       0 | 1-52
 my-app/src/components/layouts/Appshell  |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-30
 my-app/src/components/layouts/navbar    |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-55
 my-app/src/middleware                   |       0 |        0 |       0 |       0 |
  withAuth.ts                            |       0 |        0 |       0 |       0 | 1-48
 my-app/src/pages                        |       0 |        0 |       0 |       0 |
  404.tsx                                |       0 |        0 |       0 |       0 | 1-26
  _app.tsx                               |       0 |        0 |       0 |       0 | 1-19
  _document.tsx                          |       0 |        0 |       0 |       0 | 1-13
  index.tsx                              |       0 |        0 |       0 |       0 | 1-21
 my-app/src/pages/about                  |     100 |      100 |     100 |     100 |
  index.tsx                              |     100 |      100 |     100 |     100 |
 my-app/src/pages/admin                  |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-16
 my-app/src/pages/blog                   |       0 |        0 |       0 |       0 |
  [slug].tsx                             |       0 |        0 |       0 |       0 | 1-15
  index.tsx                              |       0 |        0 |       0 |       0 | 1-9
 my-app/src/pages/category               |       0 |        0 |       0 |       0 |
  [[...slug]].tsx                        |       0 |        0 |       0 |       0 | 1-18
 my-app/src/pages/editor                 |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-13
 my-app/src/pages/lapar                  |    0.98 |        0 |       0 |    0.98 |
  [id].tsx                               |       0 |        0 |       0 |       0 | 1-17
  index.tsx                              |    3.12 |        0 |       0 |    3.12 | 1-31
  server.tsx                             |       0 |        0 |       0 |       0 | 1-25
  static.tsx                             |       0 |        0 |       0 |       0 | 1-28
 my-app/src/pages/produk                 |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-34
  server.tsx                             |       0 |        0 |       0 |       0 | 1-25
  static.tsx                             |       0 |        0 |       0 |       0 | 1-28
 my-app/src/pages/produk/[produk]        |       1 |        0 |       0 |       1 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-20
  server.tsx                             |       0 |        0 |       0 |       0 | 1-29
  static.tsx                             |    1.96 |        0 |       0 |    1.96 | 1-50
 my-app/src/pages/profil                 |       0 |        0 |       0 |       0 |
  edit.tsx                               |       0 |        0 |       0 |       0 | 1-14
  index.tsx                              |       0 |        0 |       0 |       0 | 1-13
 my-app/src/pages/setting                |       0 |        0 |       0 |       0 |
  app.tsx                                |       0 |        0 |       0 |       0 | 1-9
 my-app/src/pages/shop                   |       0 |        0 |       0 |       0 |
  [[...slug]].tsx                        |       0 |        0 |       0 |       0 | 1-15
 my-app/src/pages/user                   |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-9
 my-app/src/pages/user/password          |       0 |        0 |       0 |       0 |
  index.tsx                              |       0 |        0 |       0 |       0 | 1-9
 my-app/src/utils                        |       0 |        0 |       0 |       0 |
  firebase.ts                            |       0 |        0 |       0 |       0 | 1-19
 my-app/src/utils/db                     |       0 |        0 |       0 |       0 |
  servicefirebase.ts                     |       0 |        0 |       0 |       0 | 1-139
 my-app/src/utils/swr                    |       0 |        0 |       0 |       0 |
  fetcher.ts                             |       0 |        0 |       0 |       0 | 1-2
-----------------------------------------|---------|----------|---------|---------|-------------------
Snapshot Summary
 › 1 snapshot updated from 1 test suite.

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 updated, 1 total
Time:        1.73 s
Ran all test suites.
```

Coba Jika dibuat Salah:

- Rubah menjadi toBe("About")

pemograman_framework - about.spec.tsx

```tsx
import { render } from '@testing-library/react';
import AboutPage from "@/pages/about";

describe("AboutPage", () => {
    it("renders about page correctly", () => {
        const page = render(<AboutPage />);
        expect(page.getByTestId("title").textContent).toBe("About");
        expect(page).toMatchSnapshot();
    });
})

```

Jalan kan dan Hasil:

o FAIL

Expected: "About"

Received: "  Ini Adalah Halaman About"

```
> npm run test:coverage

> my-app@0.1.0 test:coverage
> npm run test -- --coverage


> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

FAIL src/__test__/pages/about.spec.tsx
  AboutPage
    × renders about page correctly (26 ms)

  ● AboutPage › renders about page correctly

    expect(received).toBe(expected) // Object.is equality

    Expected: "About"
    Received: "Ini Adalah Halaman About"

       5 |     it("renders about page correctly", () => {
       6 |         const page = render(<AboutPage />);
       7 |         expect(page.getByTestId("title").textContent).toBe("About");
         |                                                       ^
       8 |         expect(page).toMatchSnapshot();
       9 |     });
      10 | })

      at Object.toBe (src/__test__/pages/about.spec.tsx:7:55)
```

**PRAKTIKUM 7 – TESTING PAGE DENGAN ROUTER (MOCKING)**

Kita coba untuk melakukan testing pada halaman produk

Buat file /__test__/pages/product.spec.tsx

Tambahkan kode berikut

pemograman_framework - product.spec.tsx

```tsx
import { render, screen } from '@testing-library/react';
import TampilanProduk from '@/pages/produk';

describe("Product Page", () => {
    it("renders product page correctly", () => {
        const page = render(<TampilanProduk />)
        expect(screen.getByTestId("title").textContent).toBe("Produk Page");
        expect(page).toMatchSnapshot();
    })
})
```

Ketika testing halaman Product, sering muncul error:

• NextRouter was not mounted

```
FAIL src/__test__/pages/product.spec.tsx
  ● Product Page › renders product page correctly

    NextRouter was not mounted. https://nextjs.org/docs/messages/next-router-not-mounted

      13 |     //     }
      14 |     // }, [isLogin]);
    > 15 |     const {push} = useRouter();
         |                    ^
```

Solusi: Mock Next Router

Tambahkan di file product.spec.tsx

pemograman_framework - product.spec.tsx

```tsx
import { render, screen } from '@testing-library/react';
import TampilanProduk from '@/pages/produk';

jest.mock('next/router', () => ({
    useRouter() {
        return {
            route: "/product",
            pathname: "",
            query: {},
            asPath: "",
            push: jest.fn(),
            event: {
                on: jest.fn(),
                off: jest.fn()
            },
            isReady: true,
        }
    }
}))

describe("Product Page", () => {
    it("renders product page correctly", () => {
        const page = render(<TampilanProduk />)
        expect(screen.getByTestId("title").textContent).toBe("Produk Page");
        expect(page).toMatchSnapshot();
    })
})
```

## PRAKTIKUM 8 – MENANGANI UNDEFINED DATA

Jalankan npm run test:coverage maka akan muncul error


Jika muncul error:

o Cannot read properties of undefined

o Perbaiki di komponen:

Pada file index.tsx pada folder pages/produk

pemograman_framework - index.tsx

```tsx
return (
    <div className="container mx-auto p-4">
        <TampilanProduk products={isLoading ? [] : data?.data} />
    </div>
);
```

Jalankan npm run test:coverage maka akan muncul error

Maka Solusinya perbaiki code pada file

pemograman_framework - index.tsx

```tsx
{products?.length > 0 ? (
    <>
        {products?.map((products: ProductType) => (
```

Note pastikan : comment pada code berikut pada 2 code testing

pemograman_framework - product.spec.tsx

```tsx
// expect(screen.getByTestId("title").textContent).toBe("Produk Page");
```

pemograman_framework - about.spec.tsx

```tsx
// expect(page.getByTestId("title").textContent).toBe("About");
```

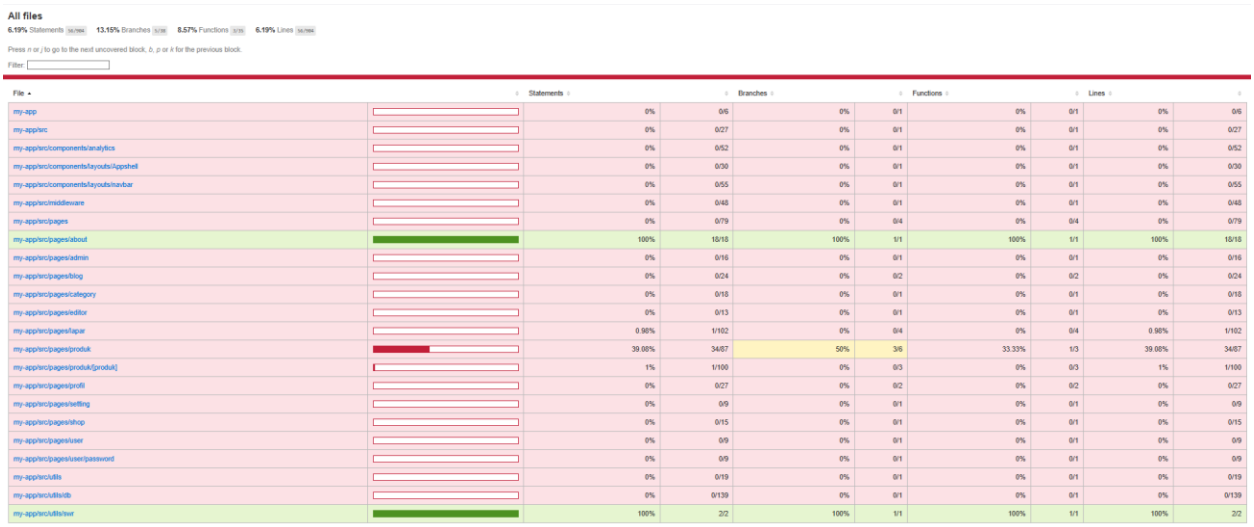
Hasil npm run test:coverage

```
> npm run test:coverage

> my-app@0.1.0 test:coverage
> npm run test -- --coverage


> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

PASS src/__test__/pages/about.spec.tsx
PASS src/__test__/pages/product.spec.tsx
```

| File | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s |
|---|---|---|---|---|---|
| All files | 6.19 | 13.15 | 8.57 | 6.19 | |
| my-app | 0 | 0 | 0 | 0 | |
| next-env.d.ts | 0 | 0 | 0 | 0 | 1-6 |
| my-app/src | 0 | 0 | 0 | 0 | |
| middleware.ts | 0 | 0 | 0 | 0 | 1-27 |
| my-app/src/components/analytics | 0 | 0 | 0 | 0 | |
| Analytics.tsx | 0 | 0 | 0 | 0 | 1-52 |
| my-app/src/components/layouts/Appshell | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-30 |
| my-app/src/components/layouts/navbar | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-55 |
| my-app/src/middleware | 0 | 0 | 0 | 0 | |
| withAuth.ts | 0 | 0 | 0 | 0 | 1-48 |
| my-app/src/pages | 0 | 0 | 0 | 0 | |
| 404.tsx | 0 | 0 | 0 | 0 | 1-26 |
| _app.tsx | 0 | 0 | 0 | 0 | 1-19 |
| _document.tsx | 0 | 0 | 0 | 0 | 1-13 |
| index.tsx | 0 | 0 | 0 | 0 | 1-21 |
| my-app/src/pages/about | 100 | 100 | 100 | 100 | |
| index.tsx | 100 | 100 | 100 | 100 | |
| my-app/src/pages/admin | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-16 |
| my-app/src/pages/blog | 0 | 0 | 0 | 0 | |
| [slug].tsx | 0 | 0 | 0 | 0 | 1-15 |
| index.tsx | 0 | 0 | 0 | 0 | 1-9 |
| my-app/src/pages/category | 0 | 0 | 0 | 0 | |
| [[...slug]].tsx | 0 | 0 | 0 | 0 | 1-18 |
| my-app/src/pages/editor | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-13 |
| my-app/src/pages/lapar | 0.98 | 0 | 0 | 0.98 | |
| [id].tsx | 0 | 0 | 0 | 0 | 1-17 |
| index.tsx | 3.12 | 0 | 0 | 3.12 | 1-31 |
| server.tsx | 0 | 0 | 0 | 0 | 1-25 |
| static.tsx | 0 | 0 | 0 | 0 | 1-28 |
| my-app/src/pages/produk | 39.08 | 50 | 33.33 | 39.08 | |
| index.tsx | 100 | 75 | 100 | 100 | 30 |
| server.tsx | 0 | 0 | 0 | 0 | 1-25 |
| static.tsx | 0 | 0 | 0 | 0 | 1-28 |
| my-app/src/pages/produk/[produk] | 1 | 0 | 0 | 1 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-20 |
| server.tsx | 0 | 0 | 0 | 0 | 1-29 |
| static.tsx | 1.96 | 0 | 0 | 1.96 | 1-50 |
| my-app/src/pages/profil | 0 | 0 | 0 | 0 | |
| edit.tsx | 0 | 0 | 0 | 0 | 1-14 |
| index.tsx | 0 | 0 | 0 | 0 | 1-13 |
| my-app/src/pages/setting | 0 | 0 | 0 | 0 | |
| app.tsx | 0 | 0 | 0 | 0 | 1-9 |
| my-app/src/pages/shop | 0 | 0 | 0 | 0 | |
| [[...slug]].tsx | 0 | 0 | 0 | 0 | 1-15 |
| my-app/src/pages/user | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-9 |
| my-app/src/pages/user/password | 0 | 0 | 0 | 0 | |
| index.tsx | 0 | 0 | 0 | 0 | 1-9 |
| my-app/src/utils | 0 | 0 | 0 | 0 | |
| firebase.ts | 0 | 0 | 0 | 0 | 1-19 |
| my-app/src/utils/db | 0 | 0 | 0 | 0 | |
| serviceFirebase.ts | 0 | 0 | 0 | 0 | 1-139 |
| my-app/src/utils/swr | 100 | 100 | 100 | 100 | |
| fetcher.ts | 100 | 100 | 100 | 100 | |

```
Test Suites: 2 passed, 2 total
Tests:       2 passed, 2 total
Snapshots:   2 passed, 2 total
Time:        2.153 s
Ran all test suites.
>
```

## ANALISIS COVERAGE



All files

6.19% Statements 13.15% Branches 8.57% Functions 6.19% Lines

Press n or j to go to the next uncovered block, b, p or k for the previous block.

Filter:

| File | | Statements | | Branches | | Functions | | Lines | |
|---|---|---|---|---|---|---|---|---|---|
| my-app | | 0% | 0/6 | 0% | 0/1 | 0% | 0/1 | 0% | 0/6 |
| my-app/src | | 0% | 0/27 | 0% | 0/1 | 0% | 0/1 | 0% | 0/27 |
| my-app/src/components/analytics | | 0% | 0/52 | 0% | 0/1 | 0% | 0/1 | 0% | 0/52 |
| my-app/src/components/layouts/Appshell | | 0% | 0/30 | 0% | 0/1 | 0% | 0/1 | 0% | 0/30 |
| my-app/src/components/layouts/navbar | | 0% | 0/55 | 0% | 0/1 | 0% | 0/1 | 0% | 0/55 |
| my-app/src/middleware | | 0% | 0/48 | 0% | 0/1 | 0% | 0/1 | 0% | 0/48 |
| my-app/src/pages | | 0% | 0/79 | 0% | 0/4 | 0% | 0/4 | 0% | 0/79 |
| my-app/src/pages/about | | 100% | 18/18 | 100% | 1/1 | 100% | 1/1 | 100% | 18/18 |
| my-app/src/pages/admin | | 0% | 0/16 | 0% | 0/1 | 0% | 0/1 | 0% | 0/16 |
| my-app/src/pages/blog | | 0% | 0/24 | 0% | 0/2 | 0% | 0/2 | 0% | 0/24 |
| my-app/src/pages/category | | 0% | 0/18 | 0% | 0/1 | 0% | 0/1 | 0% | 0/18 |
| my-app/src/pages/editor | | 0% | 0/13 | 0% | 0/1 | 0% | 0/1 | 0% | 0/13 |
| my-app/src/pages/lapar | | 0.98% | 1/102 | 0% | 0/4 | 0% | 0/4 | 0.98% | 1/102 |
| my-app/src/pages/produk | | 39.08% | 34/87 | 50% | 3/6 | 33.33% | 1/3 | 39.08% | 34/87 |
| my-app/src/pages/produk/[produk] | | 1% | 1/100 | 0% | 0/3 | 0% | 0/3 | 1% | 1/100 |
| my-app/src/pages/profil | | 0% | 0/27 | 0% | 0/2 | 0% | 0/2 | 0% | 0/27 |
| my-app/src/pages/setting | | 0% | 0/9 | 0% | 0/1 | 0% | 0/1 | 0% | 0/9 |
| my-app/src/pages/shop | | 0% | 0/15 | 0% | 0/1 | 0% | 0/1 | 0% | 0/15 |
| my-app/src/pages/user | | 0% | 0/9 | 0% | 0/1 | 0% | 0/1 | 0% | 0/9 |
| my-app/src/pages/user/password | | 0% | 0/9 | 0% | 0/1 | 0% | 0/1 | 0% | 0/9 |
| my-app/src/utils | | 0% | 0/19 | 0% | 0/1 | 0% | 0/1 | 0% | 0/19 |
| my-app/src/utils/db | | 0% | 0/139 | 0% | 0/1 | 0% | 0/1 | 0% | 0/139 |
| my-app/src/utils/swr | | 100% | 2/2 | 100% | 1/1 | 100% | 1/1 | 100% | 2/2 |

Jika dilihat aplikasi yang dibuat masih 5% jadi perlu banyak perbaikan sebelum proses deploy.

Perhatikan bagian:

- Statement
- Branch
- Function
- Lines

Contoh:

| **Metric** | **Hasil** |
|---|---|
| **Statements** | 85% |
| **Branch** | 60% |
| **Functions** | 90% |
| **Lines** | 88% |

Branch biasanya paling sulit karena perlu menguji kondisi if/else.

## STANDAR INDUSTRI

Biasanya:

- ≥ 80% → boleh production

• < 80% → harus diperbaiki

• Semua critical feature wajib dites

**TUGAS PRAKTIKUM**

**1. Buat unit test untuk:**

o Halaman Product

o 1 Komponen

Product

pemograman_framework - product.spec.tsx

```tsx
import { render, screen } from '@testing-library/react';
import TampilanProduk from '@/pages/produk';

jest.mock('next/router', () => ({
    useRouter() {
        return {
            route: "/product",
            pathname: "",
            query: {},
            asPath: "",
            push: jest.fn(),
            event: {
                on: jest.fn(),
                off: jest.fn()
            },
            isReady: true,
        }
    }
}))

describe("Product Page", () => {
    it("renders product page correctly", () => {
        const page = render(<TampilanProduk />)
        expect(screen.getByTestId("title").textContent).toBe("Produk Page");
        expect(page).toMatchSnapshot();
    })
})
```

Hero

pemograman_framework - produk-hero.spec.tsx

```tsx
import { render, screen } from '@testing-library/react';
import TampilanHeroProduk from '@/views/produk/hero';

describe('TampilanHeroProduk', () => {
    it('renders hero title correctly', () => {
        const page = render(<TampilanHeroProduk />);
        expect(screen.getByText('Produk Page').textContent).toBe('Produk Page');
        expect(page).toMatchSnapshot();
    });
});

```

Hasil

```
PASS  src/__test__/components/produk-hero.spec.tsx
PASS  src/__test__/pages/product.spec.tsx
PASS  src/__test__/pages/about.spec.tsx
```

```
Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
Snapshots:   3 passed, 3 total
Time:        2.637 s
Ran all test suites.
```

**2. Gunakan minimal:**

o 1 Snapshot test

o 1 toBe()

o 1 getByTestId()

About

pemograman_framework - index.tsx

```
11  <h1 data-testid="title">Ini Adalah Halaman About</h1>
```

Unit test about

pemograman_framework - about.spec.tsx

```
7  expect(page.getByTestId("title").textContent).toBe("Ini Adalah Halaman About");
8  expect(page).toMatchSnapshot();
```

Halaman product

pemograman_framework - index.tsx

```
17  <h1 data-testid="title" className={styles.produk__title}>Daftar Produk</h1>
```

Unit test product

pemograman_framework - product.spec.tsx

```
24  expect(screen.getByTestId("title").textContent).toBe("Daftar Produk");
25  expect(page).toMatchSnapshot();
```

Komponen Hero

pemograman_framework - hero.tsx

```
6  <div data-testid="hero-title" className="bg-gray-100 p-5 text-center text-2xl font-bold">
```

Unit test hero

pemograman_framework - produk-hero.spec.tsx

```
7  expect(screen.getByTestId("hero-title").textContent).toBe('Produk Page');
8  expect(page).toMatchSnapshot();
```

Hasil

```
PASS  src/__test__/components/produk-hero.spec.tsx
PASS  src/__test__/pages/product.spec.tsx
PASS  src/__test__/pages/about.spec.tsx
```

```
Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
Snapshots:   3 passed, 3 total
Time:        2.709 s
Ran all test suites.
```

### 3. Buat coverage minimal 50%

Modifikasi jest.config.mjs

pemograman_framework -

```
import nextJest from 'next/jest.js';

const createJestConfig = nextJest({
    dir: './',
})

const config = {
    coverageProvider: 'v8',
    testEnvironment: 'jsdom',
    modulePaths: ['<rootDir>/src/'],
    collectCoverage: true,
    collectCoverageFrom: [
        'src/pages/produk/index.tsx',
        'src/views/product/index.tsx',
        'src/views/produk/hero.tsx',
        'src/views/produk/index.tsx',
        'src/views/produk/main.tsx',
    ],
    coverageThreshold: {
        global: {
            branches: 50,
            functions: 50,
            lines: 50,
            statements: 50,
        },
    },
}

export default createJestConfig(config);
```

Hasil

```
> npm run test:coverage

> my-app@0.1.0 test:coverage
> npm run test -- --coverage


> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

PASS src/__test__/components/produk-hero.spec.tsx
PASS src/__test__/pages/product.spec.tsx
PASS src/__test__/pages/about.spec.tsx
---------------|---------|----------|---------|---------|-------------------
File           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
---------------|---------|----------|---------|---------|-------------------
All files      |   62.68 |    63.63 |      60 |   62.68 |
 pages/produk  |     100 |       75 |     100 |     100 |
  index.tsx    |     100 |       75 |     100 |     100 | 30
 views/product |   68.62 |       75 |     100 |   68.62 |
  index.tsx    |   68.62 |       75 |     100 |   68.62 | 20-35
 views/produk  |   30.61 |    33.33 |   33.33 |   30.61 |
  hero.tsx     |     100 |      100 |     100 |     100 |
  index.tsx    |       0 |        0 |       0 |       0 | 1-14
  main.tsx     |       0 |        0 |       0 |       0 | 1-20
---------------|---------|----------|---------|---------|-------------------

Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
Snapshots:   3 passed, 3 total
Time:        2.575 s
Ran all test suites.
>
```


## 4. Lakukan mocking untuk router

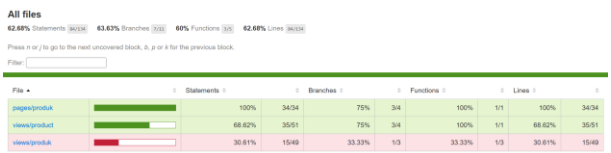
Unit test halaman produk

pemograman_framework -

```
4  jest.mock('next/router', () => ({
5      useRouter() {
6          return {
7              route: "/product",
8              pathname: "",
9              query: {},
10             asPath: "",
11             push: jest.fn(),
12             event: {
13                 on: jest.fn(),
14                 off: jest.fn()
15             },
16             isReady: true,
17         }
18     }
19 }))
```

## 5. Dokumentasikan hasil coverage

```
> npm run test:coverage

> my-app@0.1.0 test:coverage
> npm run test -- --coverage


> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

PASS src/__test__/components/produk-hero.spec.tsx
PASS src/__test__/pages/product.spec.tsx
PASS src/__test__/pages/about.spec.tsx
---------------|---------|----------|---------|---------|-------------------
File           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
---------------|---------|----------|---------|---------|-------------------
All files      |   62.68 |    63.63 |      60 |   62.68 |
 pages/produk  |     100 |       75 |     100 |     100 |
  index.tsx    |     100 |       75 |     100 |     100 | 30
 views/product |   68.62 |       75 |     100 |   68.62 |
  index.tsx    |   68.62 |       75 |     100 |   68.62 | 20-35
 views/produk  |   30.61 |    33.33 |   33.33 |   30.61 |
  hero.tsx     |     100 |      100 |     100 |     100 |
  index.tsx    |       0 |        0 |       0 |       0 | 1-14
  main.tsx     |       0 |        0 |       0 |       0 | 1-20
---------------|---------|----------|---------|---------|-------------------

Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
Snapshots:   3 passed, 3 total
Time:        2.575 s
Ran all test suites.
>
```




Note: Setelah saya modifikasi pada tugas nomor 4.

**DISKUSI & REFLEKSI**

1. Mengapa unit testing penting sebelum production?

Membantu menangkap bug lebih awal, menjaga stabilitas fitur, dan mencegah regresi saat ada perubahan kode.

2. Mengapa branch coverage sulit mencapai 100%?

Karena tidak semua jalur logika mudah dipicu, terutama kondisi edge case, error path, dan skenario yang bergantung environment tertentu.

3. Apa itu mocking?

Teknik mengganti dependensi asli (seperti API, router, atau database) dengan versi tiruan agar test fokus pada unit yang diuji.

4. Kapan snapshot test digunakan?

Snapshot test digunakan saat ingin memverifikasi output render UI tetap konsisten dari waktu ke waktu dan mendeteksi perubahan tampilan yang tidak disengaja.

5. Apakah semua file harus dites?

Tidak semua file harus dites secara langsung, tetapi file yang berisi logika penting, alur bisnis, dan komponen kritis sebaiknya memiliki cakupan test yang memadai.

**KESIMPULAN**

Dalam praktikum ini mahasiswa telah:

- ✅ Menginstal dan mengkonfigurasi Jest
- ✅ Menggunakan React Testing Library
- ✅ Membuat unit test pada pages
- ✅ Menghasilkan coverage report
- ✅ Melakukan mocking router
- ✅ Memahami pentingnya testing di industri